

Advanced Computer Networks – PCS223

Assignment 7

Name – Taranpreet Singh

Subgroup – CS4

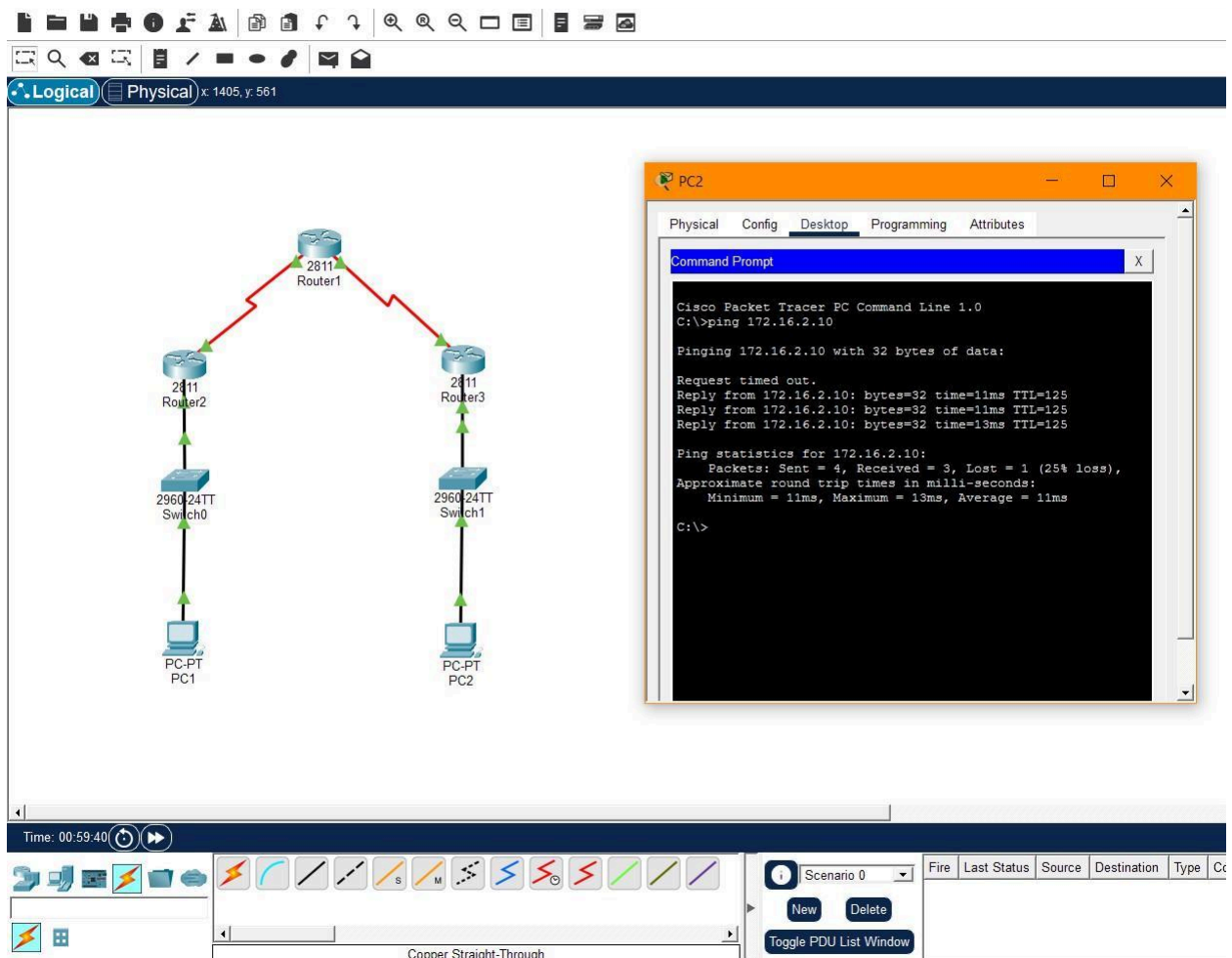
Roll No. – 8025320103

Q.1 Objective

To configure and verify routing using RIP Version 2 protocol in a network.

Problem Statement

Design and implement the given network topology in Cisco Packet Tracer with three routers (R1, R2, R3), two LANs, and PCs using the provided IP addressing scheme.



Screenshot showing network topology and successful ping from PC2 to PC1

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up

20:03:36: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Serial0/3/1 from LOADING to FULL, Loading Done

Router>enable
Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
2.2.2.2          0     FULL/ -         00:00:39    100.1.1.1      Serial0/3/0
3.3.3.3          0     FULL/ -         00:00:39    20.1.1.2       Serial0/3/1

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 1 subnets
O       10.2.2.0/24 [110/65] via 20.1.1.2, 00:08:04, Serial0/3/1
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.1.1.0/24 is directly connected, Serial0/3/1
L       20.1.1.1/32 is directly connected, Serial0/3/1
    100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       100.1.1.0/24 is directly connected, Serial0/3/0
L       100.1.1.2/32 is directly connected, Serial0/3/0
O       172.16.0.0/16 [110/65] via 100.1.1.1, 00:19:44, Serial0/3/0

Router#show ip ospf database
        OSPF Router with ID (1.1.1.1) (Process ID 1)

        Router Link States (Area 0)

Link ID      ADV Router   Age         Seq#         Checksum Link count
2.2.2.2      2.2.2.2      1184        0x80000003  0x00888c  3
1.1.1.1      1.1.1.1      512         0x80000006  0x00d931  4
3.3.3.3      3.3.3.3      490         0x80000003  0x00afab  3
Router#
```

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Press RETURN to get started.

Router>show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	0	FULL/ -	00:00:33	100.1.1.2	Serial0/3/0

Router>show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/24 is subnetted, 1 subnets
O 10.2.2.0/24 [110/129] via 100.1.1.2, 00:11:53, Serial0/3/0
20.0.0.0/24 is subnetted, 1 subnets
O 20.1.1.0/24 [110/128] via 100.1.1.2, 00:14:16, Serial0/3/0
100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 100.1.1.0/24 is directly connected, Serial0/3/0
L 100.1.1.1/32 is directly connected, Serial0/3/0
C 172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.16.0.0/16 is directly connected, FastEthernet0/0
L 172.16.2.1/32 is directly connected, FastEthernet0/0

Router>show ip database

^
% Invalid input detected at '^' marker.

Router>show ip ospf database

OSPF Router with ID (2.2.2.2) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
2.2.2.2	2.2.2.2	1415	0x80000003	0x00888c	3
1.1.1.1	1.1.1.1	743	0x80000006	0x00d931	4
3.3.3.3	3.3.3.3	721	0x80000003	0x00afab	3

Router>

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```
* incomplete command.
Router(config-router)#network 10.2.2.0 0.0.0.255 area 0
Router(config-router)#show ip ospf neighbour
      ^
% Invalid input detected at '^' marker.

Router(config-router)#exi
Router(config)#show ip ospf neighbour
      ^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip ospf neighbour
      ^
% Invalid input detected at '^' marker.

Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
1.1.1.1          0    FULL/ -         00:00:37    20.1.1.1       Serial0/3/0
Router#show ip ospf database
        OSPF Router with ID (3.3.3.3) (Process ID 1)

        Router Link States (Area 0)

Link ID          ADV Router      Age      Seq#           Checksum Link count
2.2.2.2          2.2.2.2         827      0x80000003    0x00888c 3
1.1.1.1          1.1.1.1         155      0x80000006    0x00d931 4
3.3.3.3          3.3.3.3         132      0x80000003    0x00afab 3
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.2.2.0/24 is directly connected, FastEthernet0/0
L       10.2.2.1/32 is directly connected, FastEthernet0/0
C       20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.1.1.0/24 is directly connected, Serial0/3/0
L       20.1.1.2/32 is directly connected, Serial0/3/0
C       100.0.0.0/24 is subnetted, 1 subnets
O       100.1.1.0/24 [110/128] via 20.1.1.1, 00:02:47, Serial0/3/0
O       172.16.0.0/16 [110/129] via 20.1.1.1, 00:02:47, Serial0/3/0

Router#
```

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